Cyprus University of Technology

Computer Engineering & Informatics

CEI122: Ψηφιακή Λογική

CEI_122 - Report Εργαστηρίου 6

Φοιτητές:

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Ομάδα Ν

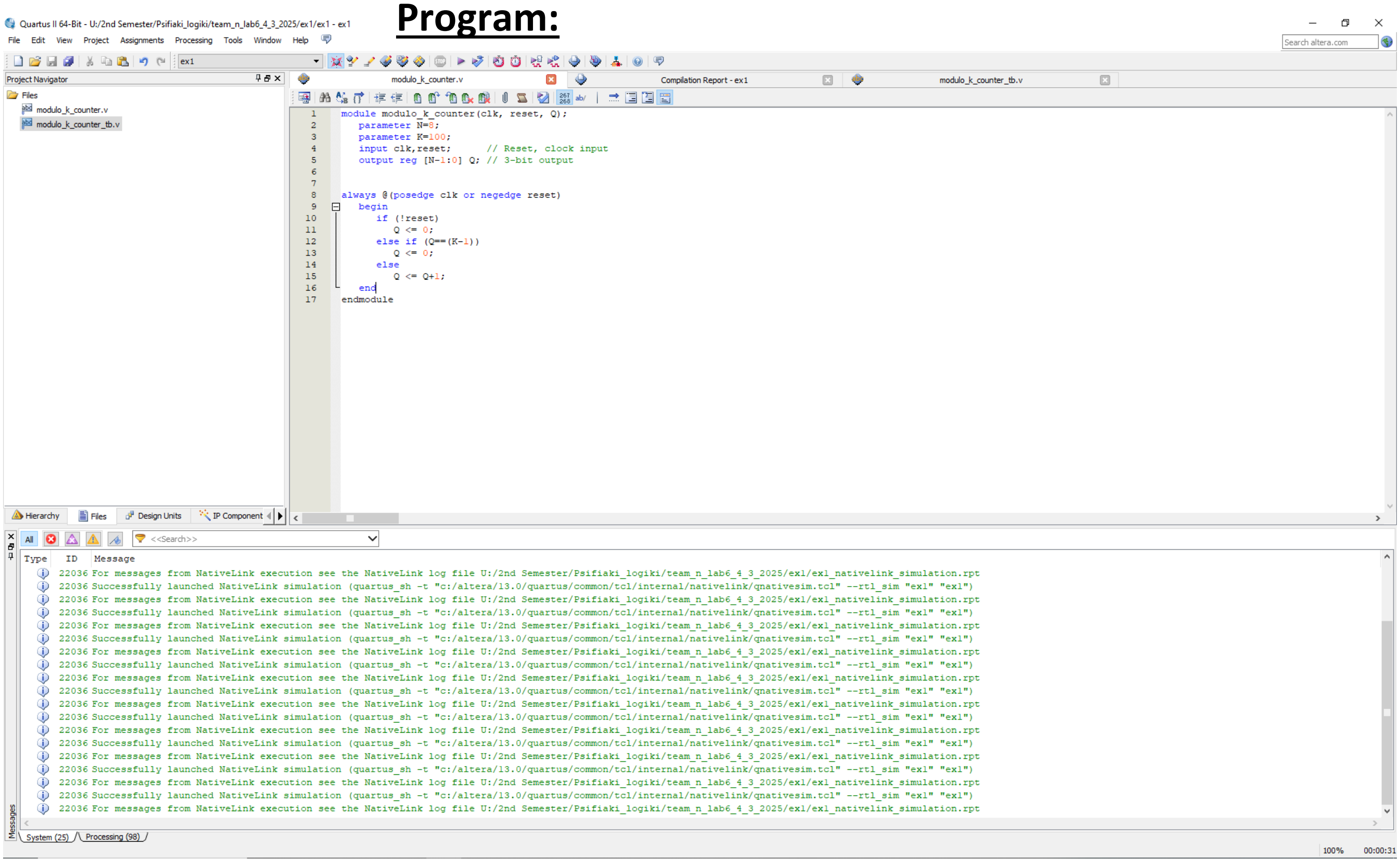
Καθηγητές:

Δρ. Κυριάκος Μ. Δεληπαράσχος

κ. Χριστόδουλος Γιαννάκου

11/03/2025

Program:


Quartus II 64-Bit - U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1 - ex1
File
Edit
View
Project
Assignments
Processing
Tools
Window
Help
Search altera.com
ex1
Project Navigator
Files
modulo_k_counter.v
modulo_k_counter_tb.v
modulo_k_counter.v
Compilation Report - ex1
modulo_k_counter_tb.v
Hierarchy
Files
Design Units
IP Component
All
<<Search>>
Messages
System (25)
Processing (98)
100%
00:00:31

```verilog
module modulo_k_counter(clk, reset, Q);
   parameter N=8;
   parameter K=100;
   input clk,reset;       // Reset, clock input
   output reg [N-1:0] Q; // 3-bit output


always @(posedge clk or negedge reset)
   begin
      if (!reset)
         Q <= 0;
      else if (Q==(K-1))
         Q <= 0;
      else
         Q <= Q+1;
   end
endmodule
```

```
Type  ID     Message
      22036 For messages from NativeLink execution see the NativeLink log file U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1_nativelink_simulation.rpt
      22036 Successfully launched NativeLink simulation (quartus_sh -t "c:/altera/13.0/quartus/common/tcl/internal/nativelink/qnativesim.tcl" --rtl_sim "ex1" "ex1")
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      22036 For messages from NativeLink execution see the NativeLink log file U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1_nativelink_simulation.rpt
```
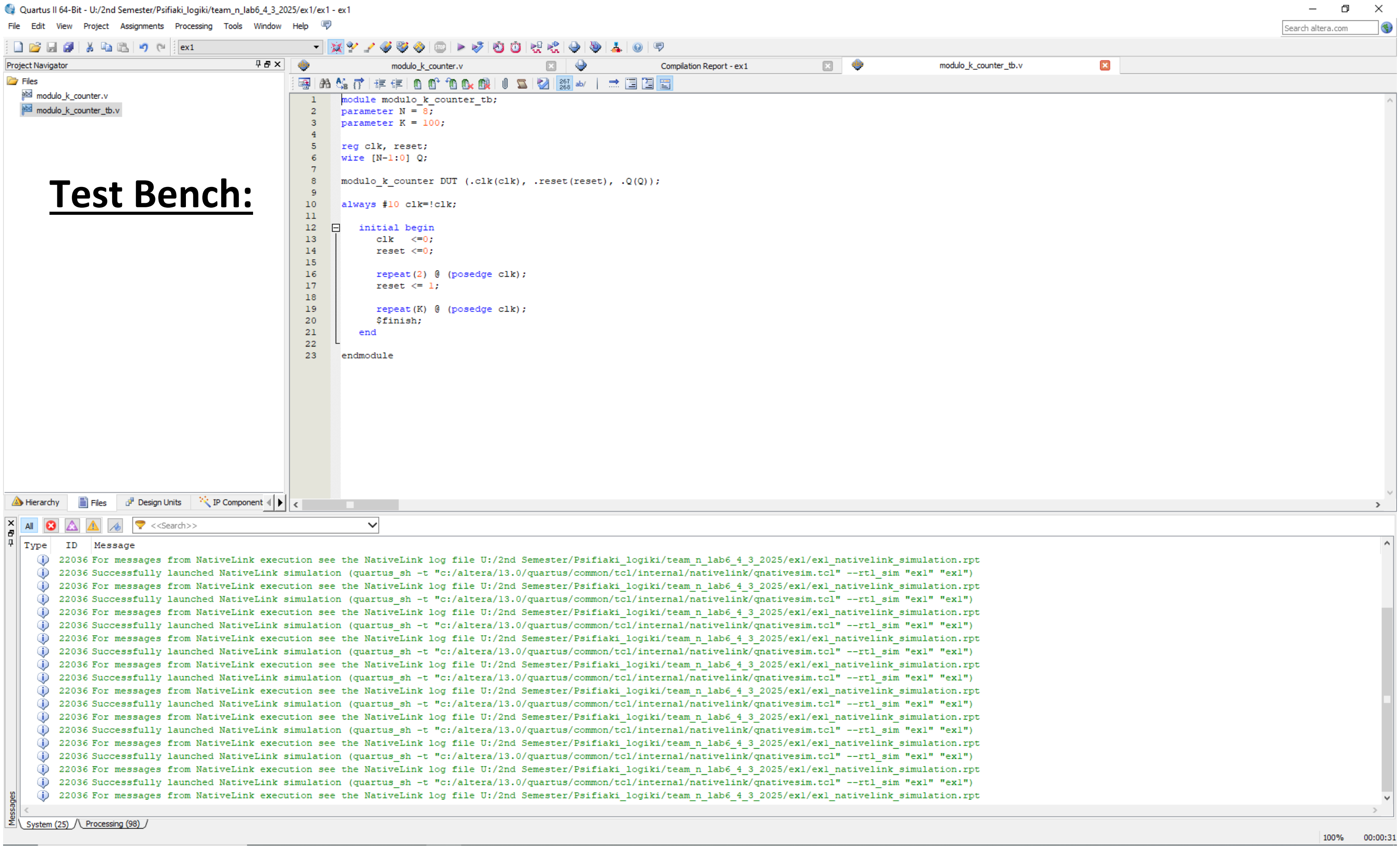

Test Bench:


Quartus II 64-Bit - U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1 - ex1
File
Edit
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modulo_k_counter_tb.v
Hierarchy
Files
Design Units
IP Component
All
<<Search>>
Messages
System (25)
Processing (98)
100%
00:00:31

```verilog
module modulo_k_counter_tb;
parameter N = 8;
parameter K = 100;

reg clk, reset;
wire [N-1:0] Q;

modulo_k_counter DUT (.clk(clk), .reset(reset), .Q(Q));

always #10 clk=!clk;

   initial begin
      clk   <=0;
      reset <=0;

      repeat(2) @ (posedge clk);
      reset <= 1;

      repeat(K) @ (posedge clk);
      $finish;
   end

endmodule
```

```
Type  ID     Message
      22036 For messages from NativeLink execution see the NativeLink log file U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1_nativelink_simulation.rpt
      22036 Successfully launched NativeLink simulation (quartus_sh -t "c:/altera/13.0/quartus/common/tcl/internal/nativelink/qnativesim.tcl" --rtl_sim "ex1" "ex1")
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      22036 For messages from NativeLink execution see the NativeLink log file U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1_nativelink_simulation.rpt
```
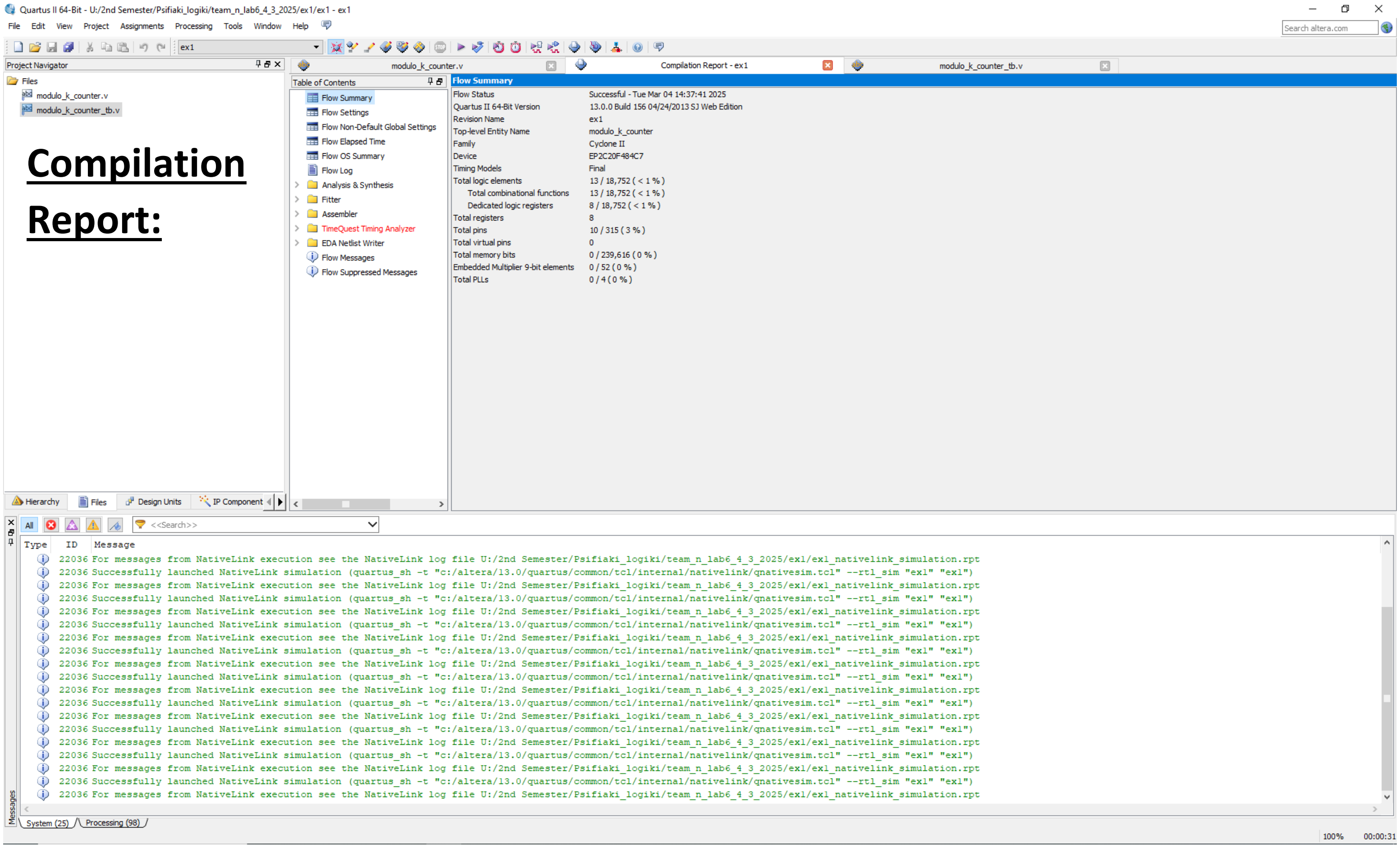

Compilation Report:


Quartus II 64-Bit - U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1 - ex1
File Edit View Project Assignments Processing Tools Window Help
Project Navigator
Files
modulo_k_counter.v
modulo_k_counter_tb.v
Compilation Report - ex1
Table of Contents
Flow Summary
Flow Settings
Flow Non-Default Global Settings
Flow Elapsed Time
Flow OS Summary
Flow Log
Analysis & Synthesis
Fitter
Assembler
TimeQuest Timing Analyzer
EDA Netlist Writer
Flow Messages
Flow Suppressed Messages
Hierarchy
Files
Design Units
IP Component
System (25)
Processing (98)
100%
00:00:31

Flow Summary

Flow Status	Successful - Tue Mar 04 14:37:41 2025
Quartus II 64-Bit Version	13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name	ex1
Top-level Entity Name	modulo_k_counter
Family	Cyclone II
Device	EP2C20F484C7
Timing Models	Final
Total logic elements	13 / 18,752 (< 1 %)
Total combinational functions	13 / 18,752 (< 1 %)
Dedicated logic registers	8 / 18,752 (< 1 %)
Total registers	8
Total pins	10 / 315 (3 %)
Total virtual pins	0
Total memory bits	0 / 239,616 (0 %)
Embedded Multiplier 9-bit elements	0 / 52 (0 %)
Total PLLs	0 / 4 (0 %)

```
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      22036 For messages from NativeLink execution see the NativeLink log file U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1_nativelink_simulation.rpt
```
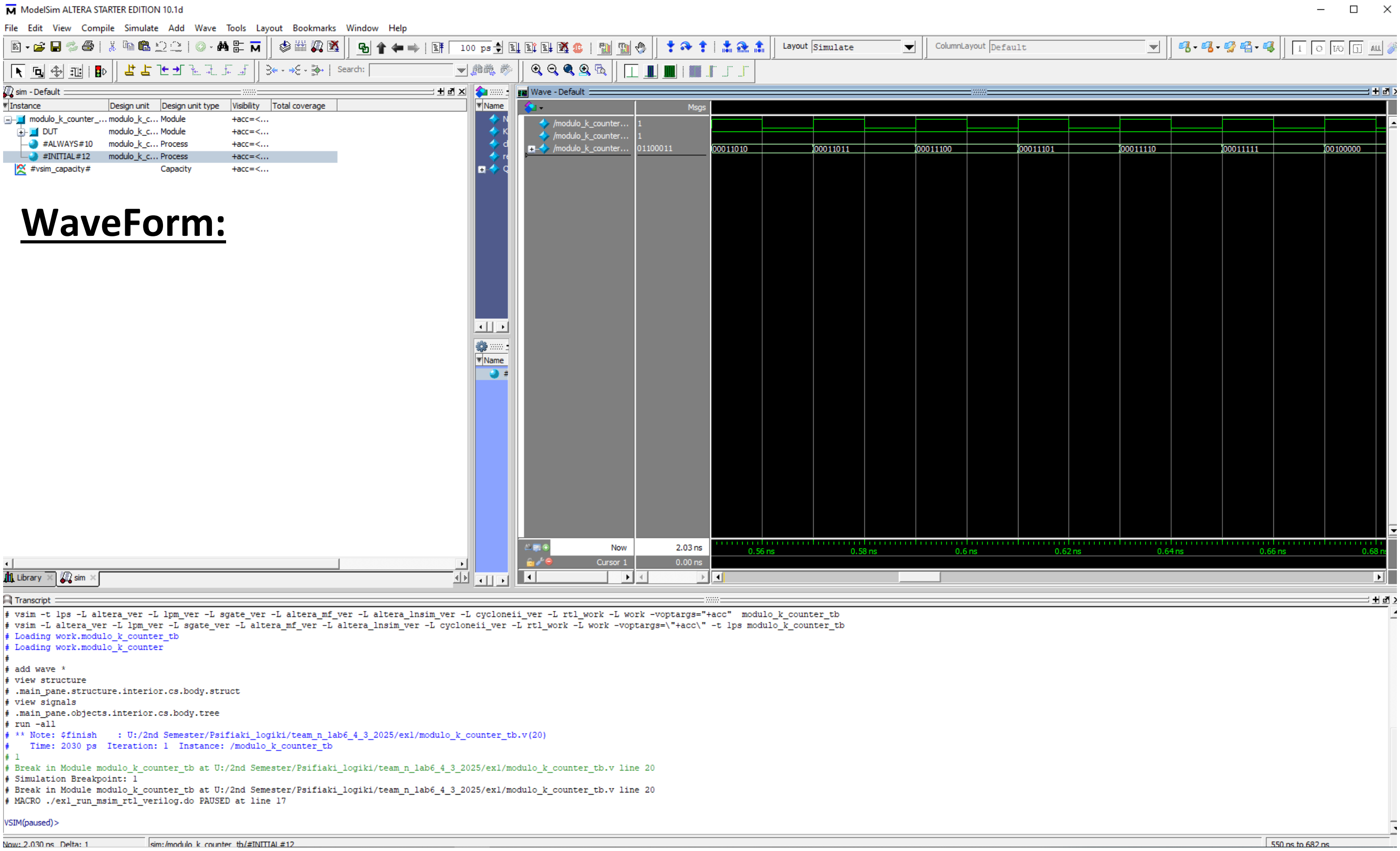
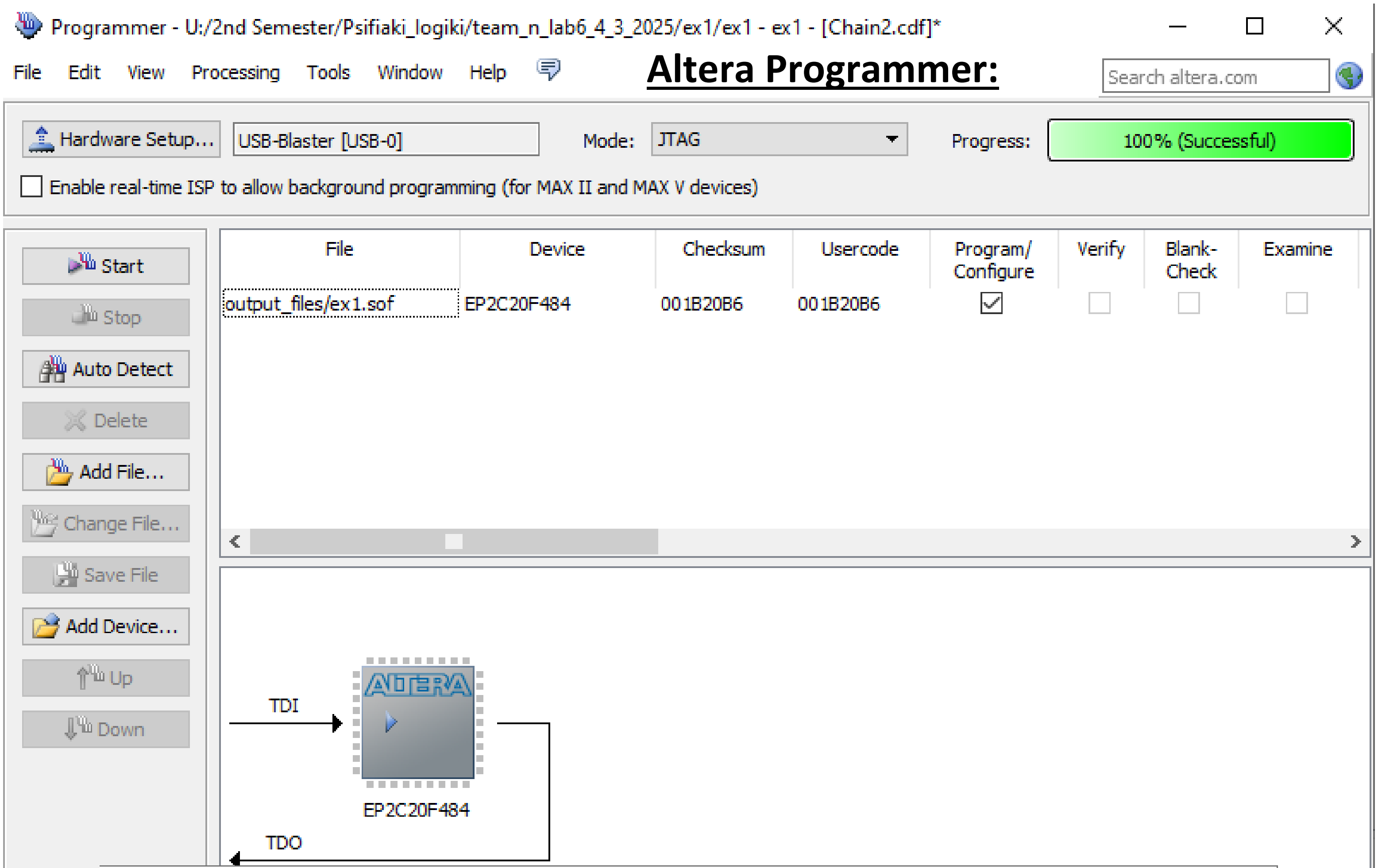

WaveForm:


ModelSim ALTERA STARTER EDITION 10.1d
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
Layout Simulate
ColumnLayout Default
sim - Default
Instance
Design unit
Design unit type
Visibility
Total coverage
modulo_k_counter_...
DUT
#ALWAYS#10
#INITIAL#12
#vsim_capacity#
Module
Process
Capacity
+acc=<...
Wave - Default
Msgs
/modulo_k_counter...
1
1
01100011
00011010
00011011
00011100
00011101
00011110
00011111
00100000
Now
2.03 ns
Cursor 1
0.00 ns
0.56 ns
0.58 ns
0.6 ns
0.62 ns
0.64 ns
0.66 ns
0.68 ns
Library
sim
Transcript
550 ns to 682 ns

```
# vsim -t 1ps -L altera_ver -L lpm_ver -L sgate_ver -L altera_mf_ver -L altera_lnsim_ver -L cycloneii_ver -L rtl_work -L work -voptargs="+acc"  modulo_k_counter_tb
# vsim -L altera_ver -L lpm_ver -L sgate_ver -L altera_mf_ver -L altera_lnsim_ver -L cycloneii_ver -L rtl_work -L work -voptargs=\"+acc\" -t 1ps modulo_k_counter_tb
# Loading work.modulo_k_counter_tb
# Loading work.modulo_k_counter
#
# add wave *
# view structure
# .main_pane.structure.interior.cs.body.struct
# view signals
# .main_pane.objects.interior.cs.body.tree
# run -all
# ** Note: $finish    : U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/modulo_k_counter_tb.v(20)
#    Time: 2030 ps  Iteration: 1  Instance: /modulo_k_counter_tb
# 1
# Break in Module modulo_k_counter_tb at U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/modulo_k_counter_tb.v line 20
# Simulation Breakpoint: 1
# Break in Module modulo_k_counter_tb at U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/modulo_k_counter_tb.v line 20
# MACRO ./ex1_run_msim_rtl_verilog.do PAUSED at line 17

VSIM(paused)>
```

```
Now: 2,030 ns  Delta: 1    sim:/modulo_k_counter_tb/#INITIAL#12
```

Altera Programmer:



Programmer - U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1 - ex1 - [Chain2.cdf]*
File
Edit
View
Processing
Tools
Window
Help
Search altera.com
Hardware Setup...
Start
Stop
Auto Detect
Delete
Add File...
Change File...
Save File
Add Device...
Up
Down
TDI
TDO
EP2C20F484

USB-Blaster [USB-0]

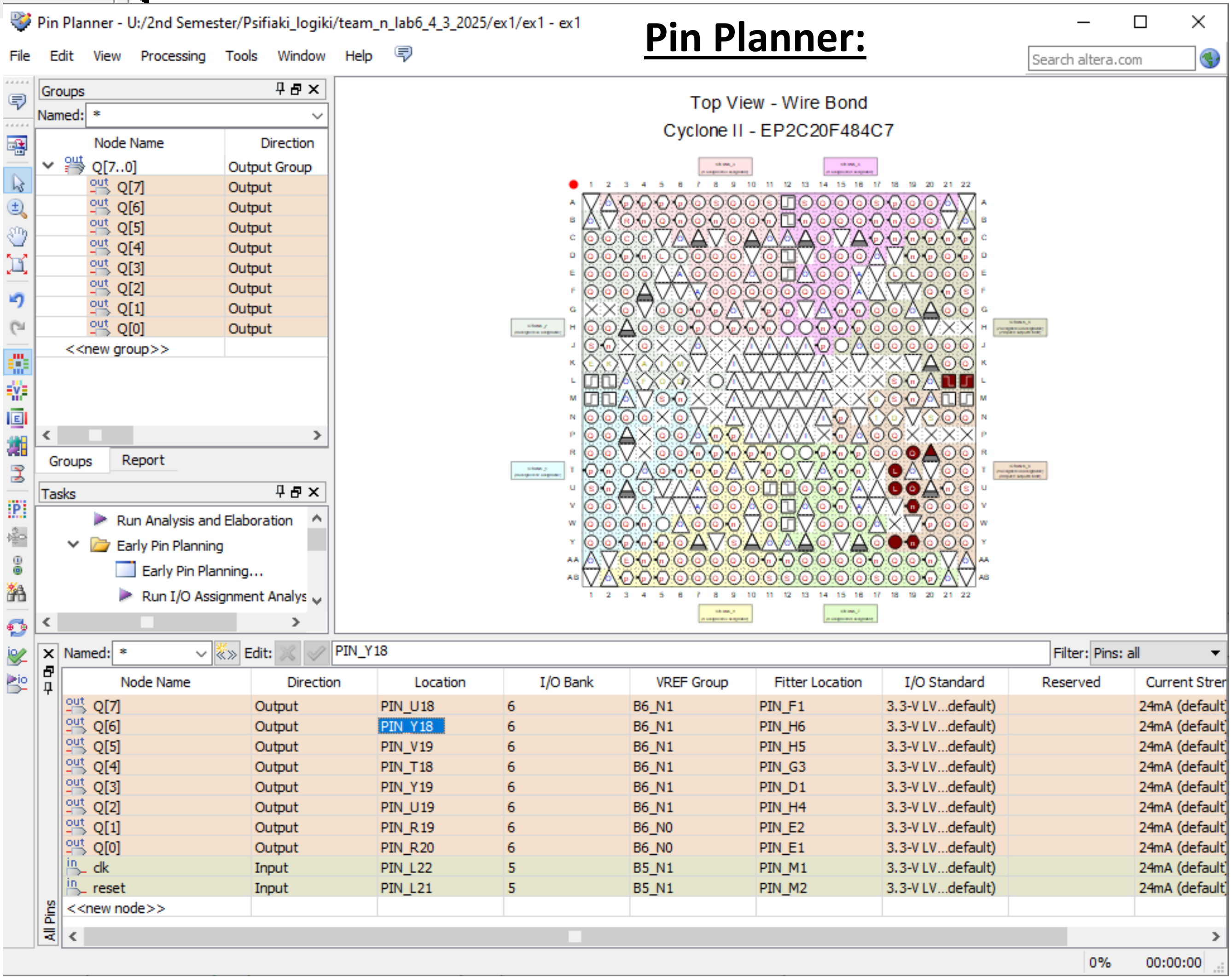
Mode: JTAG

Progress: 100% (Successful)

☐ Enable real-time ISP to allow background programming (for MAX II and MAX V devices)

File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/ex1.sof	EP2C20F484	001B20B6	001B20B6	☑	☐	☐	☐

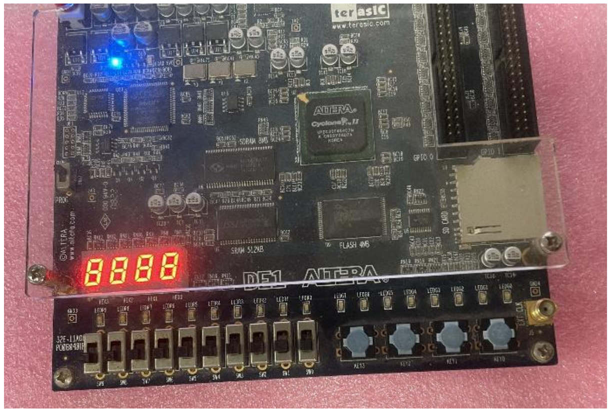
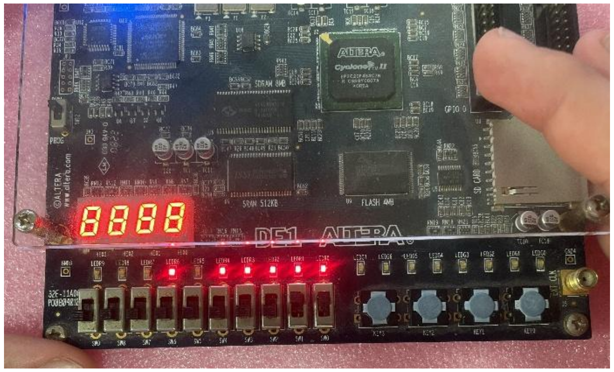
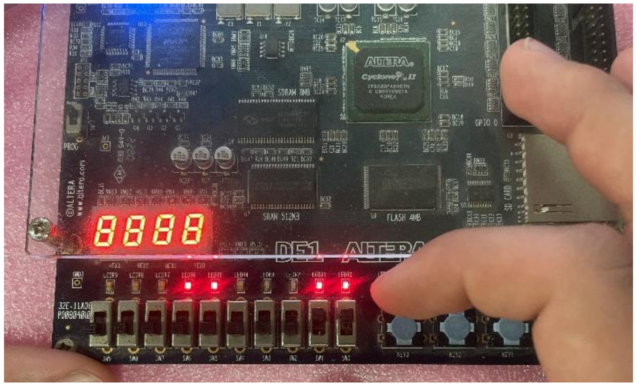
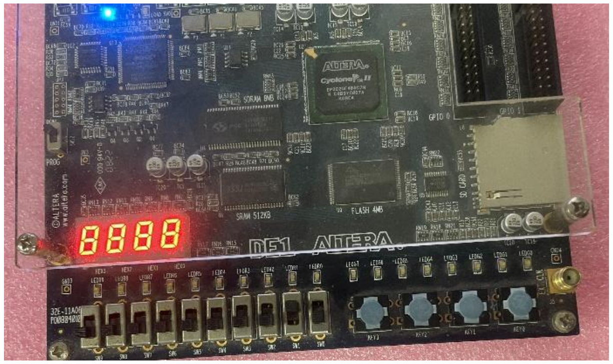
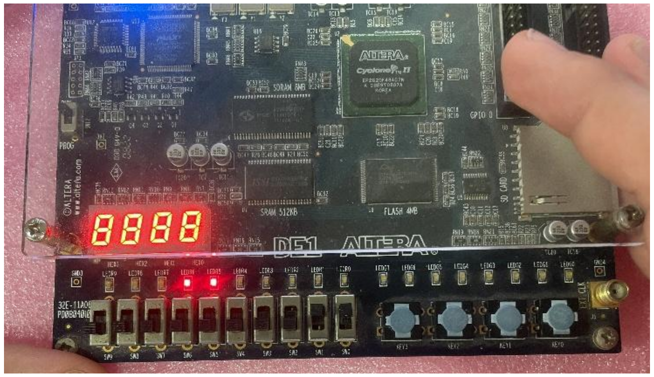
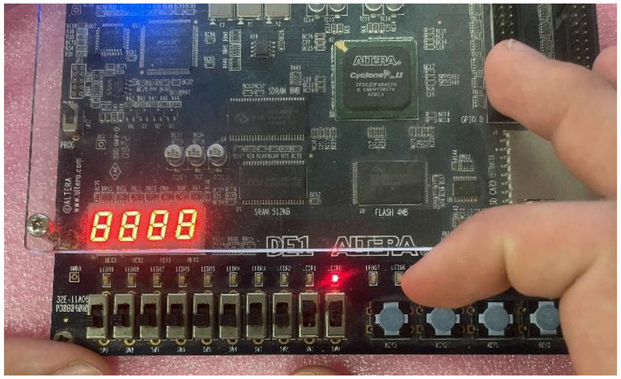
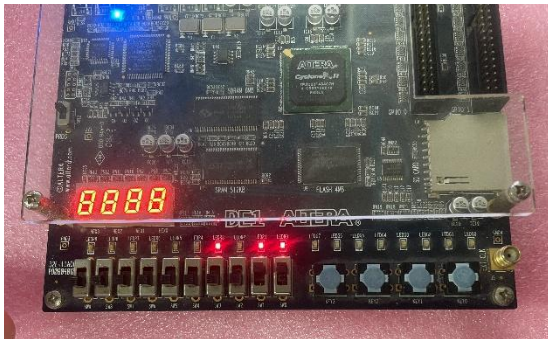
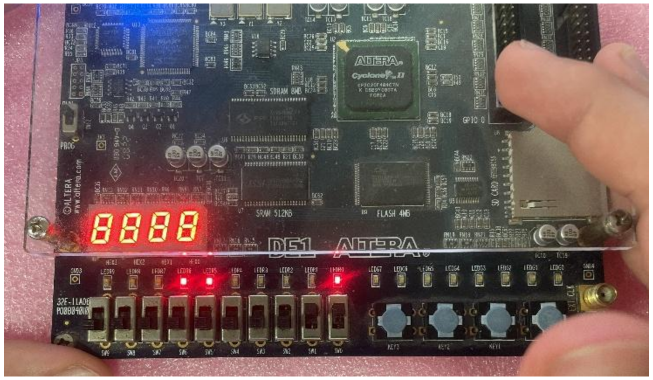
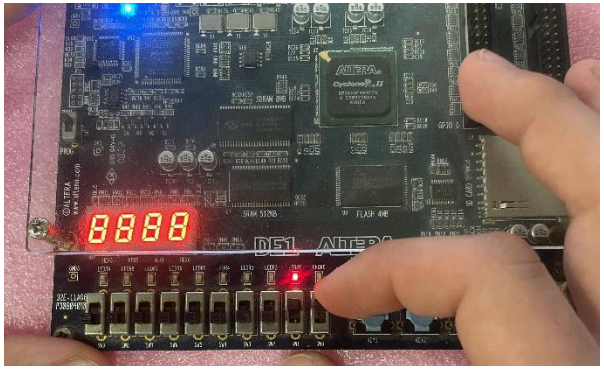
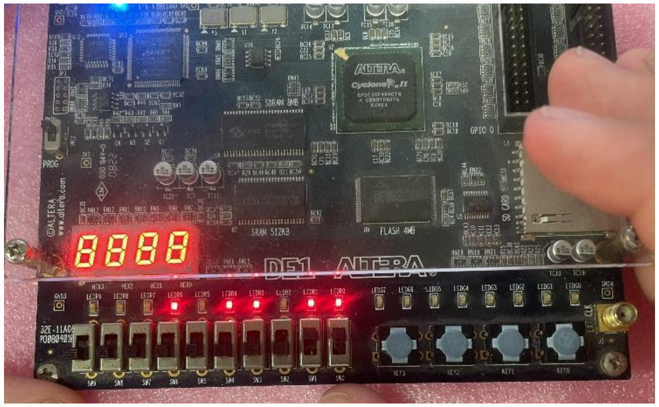
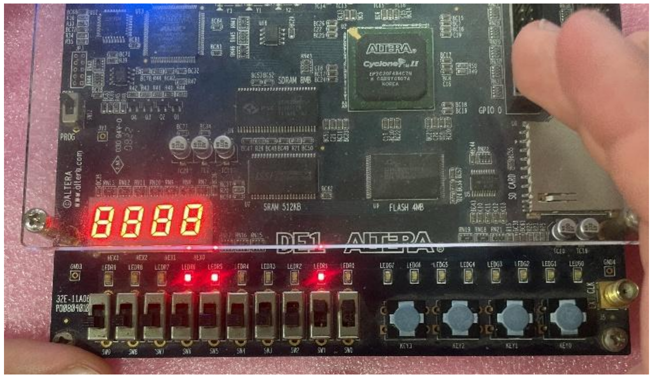
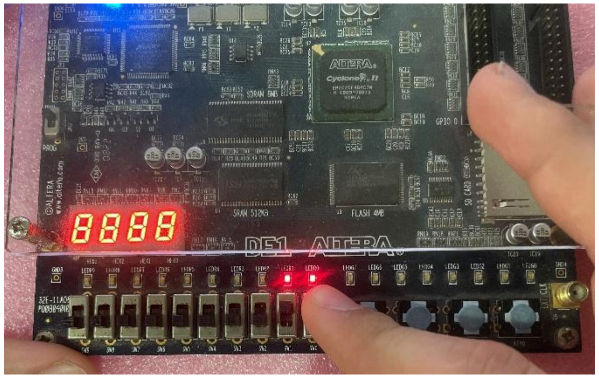
Pin Planner:



Pin Planner - U:/2nd Semester/Psifiaki_logiki/team_n_lab6_4_3_2025/ex1/ex1 - ex1
File
Edit
View
Processing
Tools
Window
Help
Search altera.com
Groups
Named: *
Groups
Report
Tasks
Run Analysis and Elaboration
Early Pin Planning
Early Pin Planning...
Run I/O Assignment Analys
Top View - Wire Bond
Cyclone II - EP2C20F484C7
All Pins
0%
00:00:00

Node Name	Direction
Q[7..0]	Output Group
Q[7]	Output
Q[6]	Output
Q[5]	Output
Q[4]	Output
Q[3]	Output
Q[2]	Output
Q[1]	Output
Q[0]	Output
<<new group>>	

Named: * Edit: PIN_Y18 Filter: Pins: all

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Stre
Q[7]	Output	PIN_U18	6	B6_N1	PIN_F1	3.3-V LV...default)		24mA (default
Q[6]	Output	PIN_Y18	6	B6_N1	PIN_H6	3.3-V LV...default)		24mA (default
Q[5]	Output	PIN_V19	6	B6_N1	PIN_H5	3.3-V LV...default)		24mA (default
Q[4]	Output	PIN_T18	6	B6_N1	PIN_G3	3.3-V LV...default)		24mA (default
Q[3]	Output	PIN_Y19	6	B6_N1	PIN_D1	3.3-V LV...default)		24mA (default
Q[2]	Output	PIN_U19	6	B6_N1	PIN_H4	3.3-V LV...default)		24mA (default
Q[1]	Output	PIN_R19	6	B6_N0	PIN_E2	3.3-V LV...default)		24mA (default
Q[0]	Output	PIN_R20	6	B6_N0	PIN_E1	3.3-V LV...default)		24mA (default
clk	Input	PIN_L22	5	B5_N1	PIN_M1	3.3-V LV...default)		24mA (default
reset	Input	PIN_L21	5	B5_N1	PIN_M2	3.3-V LV...default)		24mA (default
<<new node>>